

PRAKTIKUM SISTEM OPERASI

MODUL 15

Keamanan Linux



Oleh:

Jauhar Fajar Zuhair

2311104072

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

JURNAL

1. Soal 1

```
Q administrator@Arch:~/jauharfz_2311104072

~ via C v16.1.1-gcc
> sha256sum /etc/passwd
5ef47f56d83bdee79fbf6964767979bad5ea1beb4e080e2d8841a5b72e14c98f /etc/passwd

~ via C v16.1.1-gcc
> sha512sum /etc/passwd
4c94859a78492f664d4036791eb19673dd962d4b6561a0f7f8aca193be407db8bc2cc3fdc3547c1fada25276948980e9d5ef4e5e272faaf76bbd8c0895a3f464 /etc/passwd

~ via C v16.1.1-gcc
> md5sum /etc/passwd
dbce6d132f403153dfac2e36120c7f22 /etc/passwd

~ via C v16.1.1-gcc
> cp /etc/passwd /home/endeminator/jauharfz_2311104072/test_0.txt

~ via C v16.1.1-gcc
> cd jauharfz_2311104072/

~/jauharfz_2311104072
> sha256sum test_0.txt
5ef47f56d83bdee79fbf6964767979bad5ea1beb4e080e2d8841a5b72e14c98f test_0.txt

~/jauharfz_2311104072
> sha512sum test_0.txt
4c94859a78492f664d4036791eb19673dd962d4b6561a0f7f8aca193be407db8bc2cc3fdc3547c1fada25276948980e9d5ef4e5e272faaf76bbd8c0895a3f464 test_0.txt

~/jauharfz_2311104072
> md5sum test_0.txt
dbce6d132f403153dfac2e36120c7f22 test_0.txt

~/jauharfz_2311104072
> mv test_0.txt file_0.txt

~/jauharfz_2311104072
> sha256sum file_0.txt
5ef47f56d83bdee79fbf6964767979bad5ea1beb4e080e2d8841a5b72e14c98f file_0.txt

~/jauharfz_2311104072
> sha512sum file_0.txt
4c94859a78492f664d4036791eb19673dd962d4b6561a0f7f8aca193be407db8bc2cc3fdc3547c1fada25276948980e9d5ef4e5e272faaf76bbd8c0895a3f464 file_0.txt

~/jauharfz_2311104072
> md5sum file_0.txt
dbce6d132f403153dfac2e36120c7f22 file_0.txt
```

2. Soal 2

```
Q administrator@Arch:~/jauharfz_2311104072

Saving to: 'test_1.txt'

test_1.txt [ <=> ] 388.32K --.-KB/s in 0.1s

2026-06-18 03:51:12 (2.78 MB/s) - 'test_1.txt' saved [397637]

~/jauharfz_2311104072 took 2s
> sha256sum test_1.txt
80164829d7be76a52ac8160bff81cbf2bae6d4304f1fd43baa03d41f468793ee test_1.txt

~/jauharfz_2311104072
> sha512sum test_1.txt
d5e8b09df9cd7cb93444e761f24da2bd05e6ddbfc4a60a19b64ce5cc9116f4ab3bbd828c79daa9cfed672753203ea007482ef033a95d2a5a4a1c54d48f62ef test_1.txt

~/jauharfz_2311104072
> md5sum test_1.txt
6b476a9f699497ced5524c4a073e4812 test_1.txt

~/jauharfz_2311104072
> nvim test_1.txt

~/jauharfz_2311104072 took 55s
> sha256sum test_1.txt
5b42ef1ac89c5bc48553fbb388df1ba3fe5f8073e6c606a1341159cb09ec422b test_1.txt

~/jauharfz_2311104072
> sha512sum test_1.txt
9bdb4b4cfd12397afcd2ad8f0d32173b72db9f245618124653a15a5d2c90eda0a7f7be776f05a4dd17af2d759bdb3b33cb000670382e9ed2a957ae7981de45ec3 test_1.txt

~/jauharfz_2311104072
> md5sum test_1.txt
39d08e040fcdab06bc9ad791c50fbac test_1.txt

~/jauharfz_2311104072
> █
```

3. Soal 3

```
Q administrator@Arch:~/jauharfz_2311104072

> sha512sum test_1.txt
9bdb4b4cfd12397afc2ad8f0d32173b72db9f245618124653a15a5d2c90eda0a7f7b76f05a4dd17af2d759bdb3b33cb000670382e9ed2a957ae7981de45ec3 test_1.txt

~/jauharfz_2311104072
> md5sum test_1.txt
39d08e040fcdabab6ebc9ad791c50fbac test_1.txt

~/jauharfz_2311104072
> sha256sum test_1.doc
1505a649cc8022fa03eeabb178499dfb2ae32a3d3311b9a1a5626a68e4f00c43 test_1.doc

~/jauharfz_2311104072
> sha512sum test_1.doc
bb9b172949e150afb4d4f1213fe71b7bff3d16a08fe83d4149e9bb3b2cb68e6c14281a59cb66aea58bf520fac02ae32d79032bbb072ad41584bfec28641a3727 test_1.doc

~/jauharfz_2311104072
> md5sum test_1.doc
c78dda37d12312cde23237a55c44751 test_1.doc

~/jauharfz_2311104072
> nvim test_1.doc

~/jauharfz_2311104072 took 38s
> sha256sum test_1.doc
715068a0876b02070783545f98ab36f1b05c0e950bad9b14c4556e450cf96024 test_1.doc

~/jauharfz_2311104072
> sha512sum test_1.doc
309ab0166580373262ef5d9644c2fffb2b11e7b570e06c7d66b5090391c25cee8d6184f87bc90b6079699090bfaaf645b61f09a252583859466a21453676be94e test_1.doc

~/jauharfz_2311104072
> md5sum test_1.doc
7b5b1fdd4007164dc3e128d18557c812 test_1.doc

~/jauharfz_2311104072
> █
```

4. Soal 4abcd

```
Q administrator@Arch:~/jauharfz_2311104072

~/jauharfz_2311104072
> cp test_1.doc test_1.txt file_0.txt normal_directory/

~/jauharfz_2311104072
> ls -la normal_directory/
total 44
drwx----- 2 administrator endministrator 4096 Jun 18 04:07 .
drwxr-xr-x 4 administrator endministrator 4096 Jun 18 04:00 ..
-rw-r--r-- 1 administrator endministrator 2316 Jun 18 04:07 file_0.txt
-rw-r--r-- 1 administrator endministrator 27137 Jun 18 04:07 test_1.doc
-rw-r--r-- 1 administrator endministrator 445 Jun 18 04:07 test_1.txt

~/jauharfz_2311104072
> ls -la encrypted_directory/
total 48
drwx----- 2 administrator endministrator 4096 Jun 18 04:07 .
drwxr-xr-x 4 administrator endministrator 4096 Jun 18 04:00 ..
-rw-r--r-- 1 administrator endministrator 453 Jun 18 04:07 8nMswmoGuMvrHkzcdnb7wboz
-rw-r--r-- 1 administrator endministrator 1298 Jun 18 04:00 .encfs6.xml
-rw-r--r-- 1 administrator endministrator 2324 Jun 18 04:07 tBh0TJdYd,jVGDMZ0TJl0Kpm
-rw-r--r-- 1 administrator endministrator 27145 Jun 18 04:07 zozD1yJOptdRaZ7NH7J4zE-F

~/jauharfz_2311104072
> rm encrypted_directory/tBh0TJdYd,jVGDMZ0TJl0Kpm

~/jauharfz_2311104072
> ls -la normal_directory/
total 40
drwx----- 2 administrator endministrator 4096 Jun 18 04:09 .
drwxr-xr-x 4 administrator endministrator 4096 Jun 18 04:00 ..
-rw-r--r-- 1 administrator endministrator 27137 Jun 18 04:07 test_1.doc
-rw-r--r-- 1 administrator endministrator 445 Jun 18 04:07 test_1.txt

~/jauharfz_2311104072
> █
```

```

Q endadministrator@Arch:~/jauharfz_2311104072
:: Running post-transaction hooks...
(1/1) Arming ConditionNeedsUpdate...

~/jauharfz_2311104072 took 4s
> encfs ~/jauharfz_2311104072/encrypted_directory ~/jauharfz_2311104072/normal_directory
The directory "/home/endadministrator/jauharfz_2311104072/encrypted_directory/" does not exist. Should it be created? (y,N) y
The directory "/home/endadministrator/jauharfz_2311104072/normal_directory/" does not exist. Should it be created? (y,N) y
Creating new encrypted volume.
Please choose from one of the following options:
  enter "x" for expert configuration mode,
  enter "p" for pre-configured paranoia mode,
  anything else, or an empty line will select standard mode.
?>

Standard configuration selected.

Configuration finished. The filesystem to be created has
the following properties:
Filesystem cipher: "ssl/aes", version 3:0:2
Filename encoding: "nameio/block", version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
Filenames encoded using IV chaining mode.
File holes passed through to ciphertext.

Now you will need to enter a password for your filesystem.
You will need to remember this password, as there is absolutely
no recovery mechanism. However, the password can be changed
later using encfsctl.

New Encfs Password:
Verify Encfs Password:

~/jauharfz_2311104072 took 22s
> █

Q endadministrator@Arch:~/jauharfz_2311104072
-rw-r--r-- 1 endadministrator endadministrator 1298 Jun 18 04:00 .encfs6.xml
-rw-r--r-- 1 endadministrator endadministrator 2324 Jun 18 04:07 tBh0TJdYd,jVGDMZ0TJl0Kpm
-rw-r--r-- 1 endadministrator endadministrator 27145 Jun 18 04:07 zozDiyJ0ptdRaZ7NH7J4zE-F

~/jauharfz_2311104072
> rm encrypted_directory/tBh0TJdYd,jVGDMZ0TJl0Kpm

~/jauharfz_2311104072
> ls -la normal_directory/
total 40
drwx----- 2 endadministrator endadministrator 4096 Jun 18 04:09 .
drwxr-xr-x 4 endadministrator endadministrator 4096 Jun 18 04:09 ..
-rw-r--r-- 1 endadministrator endadministrator 27137 Jun 18 04:07 test_1.doc
-rw-r--r-- 1 endadministrator endadministrator 445 Jun 18 04:07 test_1.txt

~/jauharfz_2311104072
> fusermount -u normal_directory/

~/jauharfz_2311104072
> ls -la normal_directory/
total 8
drwx----- 2 endadministrator endadministrator 4096 Jun 18 04:09 .
drwxr-xr-x 4 endadministrator endadministrator 4096 Jun 18 04:09 ..

~/jauharfz_2311104072
> ls -la encrypted_directory/
total 44
drwx----- 2 endadministrator endadministrator 4096 Jun 18 04:09 .
drwxr-xr-x 4 endadministrator endadministrator 4096 Jun 18 04:09 ..
-rw-r--r-- 1 endadministrator endadministrator 453 Jun 18 04:07 8nMswmoGuMvrHkzcdnb7wboz
-rw-r--r-- 1 endadministrator endadministrator 1298 Jun 18 04:00 .encfs6.xml
-rw-r--r-- 1 endadministrator endadministrator 27145 Jun 18 04:07 zozDiyJ0ptdRaZ7NH7J4zE-F

~/jauharfz_2311104072
> mkdir -p random_directory

```

5. Soal 4e+5a

```
Q administrator@Arch:~/jauharfz_2311104072
> encfs ~/jauharfz_2311104072/encrypted_directory ~/jauharfz_2311104072/random_directory
EncFS Password:

~/jauharfz_2311104072 took 3s
> gpg --gen-key
gpg (GnuPG) 2.4.9; Copyright (C) 2025 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Note: Use "gpg --full-generate-key" for a full featured key generation dialog.

GnuPG needs to construct a user ID to identify your key.

Real name: jauharfz
Email address: jauharf14@gmail.com
You selected this USER-ID:
"jauharfz <jauharf14@gmail.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit?
Change (N)ame, (E)mail, or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: directory '/home/endadministrator/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/endadministrator/.gnupg/openpgp-revocs.d/FDB2FF6A44B4751AD928B08F53021BF105F0C2F.rev'
public and secret key created and signed.

pub   ed25519 2026-06-17 [SC] [expires: 2029-06-16]
       FDB2FF6A44B4751AD928B08F53021BF105F0C2F
uid           jauharfz <jauharf14@gmail.com>
sub   cv25519 2026-06-17 [E] [expires: 2029-06-16]
```

6. Soal 5b

```
Q administrator@Arch:~/jauharfz_2311104072
~/jauharfz_2311104072 took 22s
> gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2029-06-16
[keyboard]
-----
pub   rsa4096 2024-04-09 [SC] [expired: 2026-04-09]
       013A0127AC9C65B34FFA62526C1009B693975393
uid           [ expired] Carlos Garcia Campos <cgarcia@igalia.com>

pub   dsa1024 2008-06-02 [SCA] [expires: 2027-03-16]
       5AA3BC334FD7E3369E7C77B291C559DBE4C9123B
uid           [ unknown] Adrián Pérez de Castro <aperez@igalia.com>
uid           [ unknown] Adrián Pérez de Castro (personal) <adrian@perezdecastro.org>
sub   elg4096 2008-06-02 [E] [expires: 2027-03-16]

pub   rsa4096 2025-11-21 [SC] [expires: 2027-02-14]
       E1096BCBFF6D418796DE78515384CE82BA52C83A
uid           [ unknown] Spotify Public Repository Signing Key <tux@spotify.com>

pub   ed25519 2026-06-17 [SC] [expires: 2029-06-16]
       FDB2FF6A44B4751AD928B08F53021BF105F0C2F
uid           [ultimate] jauharfz <jauharf14@gmail.com>
sub   cv25519 2026-06-17 [E] [expires: 2029-06-16]

~/jauharfz_2311104072
> gpg --list-keys jauharf14@gmail.com
pub   ed25519 2026-06-17 [SC] [expires: 2029-06-16]
       FDB2FF6A44B4751AD928B08F53021BF105F0C2F
uid           [ultimate] jauharfz <jauharf14@gmail.com>
sub   cv25519 2026-06-17 [E] [expires: 2029-06-16]
```

7. Soal 5c-f

```
Q administrator@Arch:~/jauharfz_2311104072

> gpg --list-keys jauharf14@gmail.com
pub  ed25519 2026-06-17 [SC] [expires: 2029-06-16]
    FDB2FF6A44B4751AD928808F53021BF105F0C2F
uid      [ultimate] jauharfz <jauharf14@gmail.com>
sub  cv25519 2026-06-17 [E] [expires: 2029-06-16]

~/jauharfz_2311104072
> gpg --armor --export jauharf14@gmail.com > mypublic_key.asc

~/jauharfz_2311104072
> mv mypublic_key.asc 2311104072.asc

~/jauharfz_2311104072
> gpg --import 2311104071.asc
gpg: can't open '2311104071.asc': No such file or directory
gpg: Total number processed: 0

~/jauharfz_2311104072
> gpg --import 2311104072.asc
gpg: key F53021BF105F0C2F: "jauharfz <jauharf14@gmail.com>" not changed
gpg: Total number processed: 1
gpg:      unchanged: 1

~/jauharfz_2311104072
> gpg --encrypt --armor -r jauharf14@gmail.com file_0.txt

~/jauharfz_2311104072
> gpg file_0.txt.asc
gpg: WARNING: no command supplied. Trying to guess what you mean ...
gpg: encrypted with cv25519 key, ID 2BC9601ABC6B4161, created 2026-06-17
    "jauharfz <jauharf14@gmail.com>"
File 'file_0.txt' exists. Overwrite? (y/N) y

~/jauharfz_2311104072 took 6s
> █
```